MSG No. 66.498 C_ccm [Type IV, orthorhombic]

Table 1: Wyckoff site: `4a`, site symmetry: `m'm'm`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,4,5,8,18,19,22,23]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,6,7,17,20,21,24]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,12,13,16,26,27,30,31]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,14,15,25,28,29,32]`

Table 2: Wyckoff site: `4b`, site symmetry: `m'm'm`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,4,5,8,18,19,22,23]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,6,7,17,20,21,24]`
3	$\left[0,\ \frac{1}{2},\ 0\right]$	`[9,12,13,16,26,27,30,31]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,14,15,25,28,29,32]`

Table 3: Wyckoff site: `4c`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,21,22,23,24]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8,17,18,19,20]`
3	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[9,10,11,12,29,30,31,32]`
4	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14,15,16,25,26,27,28]`

Table 4: Wyckoff site: `4d`, site symmetry: `m'm'm'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,21,22,23,24]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[5,6,7,8,17,18,19,20]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[9,10,11,12,29,30,31,32]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[13,14,15,16,25,26,27,28]`

Table 5: Wyckoff site: `8e`, site symmetry: `..2/m`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ 0\right]$	`[1,8,12,13]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}\right]$	`[2,7,11,14]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[3,6,10,15]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ 0\right]$	`[4,5,9,16]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}\right]$	`[17,24,28,29]`
6	$\left[\frac{1}{4},\ \frac{3}{4},\ 0\right]$	`[18,23,27,30]`
7	$\left[\frac{3}{4},\ \frac{1}{4},\ 0\right]$	`[19,22,26,31]`

8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[20,21,25,32]

Table 6: Wyckoff site: 8f, site symmetry: ..2/m'

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[1,12,24,29]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[2,11,23,30]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[3,10,22,31]
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[4,9,21,32]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[5,16,20,25]
6	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[6,15,19,26]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[7,14,18,27]
8	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[8,13,17,28]

Table 7: Wyckoff site: 8g, site symmetry: 2'm'm

No.	position	mapping
1	$[x, 0, 0]$	[1,8,18,23]
2	$\left[x, 0, \frac{1}{2}\right]$	[2,7,17,24]
3	$\left[-x, 0, \frac{1}{2}\right]$	[3,6,20,21]
4	$[-x, 0, 0]$	[4,5,19,22]
5	$\left[x+\frac{1}{2}, \frac{1}{2}, 0\right]$	[9,16,26,31]
6	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[10,15,25,32]
7	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[11,14,28,29]
8	$\left[\frac{1}{2}-x, \frac{1}{2}, 0\right]$	[12,13,27,30]

Table 8: Wyckoff site: 8h, site symmetry: 2m'm'

No.	position	mapping
1	$\left[x, 0, \frac{1}{4}\right]$	[1,2,23,24]
2	$\left[-x, 0, \frac{1}{4}\right]$	[3,4,21,22]
3	$\left[-x, 0, \frac{3}{4}\right]$	[5,6,19,20]
4	$\left[x, 0, \frac{3}{4}\right]$	[7,8,17,18]
5	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	[9,10,31,32]
6	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{4}\right]$	[11,12,29,30]
7	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{3}{4}\right]$	[13,14,27,28]
8	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	[15,16,25,26]

Table 9: Wyckoff site: 8i, site symmetry: m'2'm

No.	position	mapping
1	$[0, y, 0]$	[1,8,19,22]
2	$\left[0, -y, \frac{1}{2}\right]$	[2,7,20,21]
3	$\left[0, y, \frac{1}{2}\right]$	[3,6,17,24]

continued ...

Table 9

No.	position	mapping
4	$[0, -y, 0]$	[4,5,18,23]
5	$\left[\frac{1}{2}, y+\frac{1}{2}, 0\right]$	[9,16,27,30]
6	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{1}{2}\right]$	[10,15,28,29]
7	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{2}\right]$	[11,14,25,32]
8	$\left[\frac{1}{2}, \frac{1}{2}-y, 0\right]$	[12,13,26,31]

Table 10: Wyckoff site: 8j, site symmetry: m'2m'

No.	position	mapping
1	$\left[0, y, \frac{1}{4}\right]$	[1,3,22,24]
2	$\left[0, -y, \frac{1}{4}\right]$	[2,4,21,23]
3	$\left[0, -y, \frac{3}{4}\right]$	[5,7,18,20]
4	$\left[0, y, \frac{3}{4}\right]$	[6,8,17,19]
5	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{1}{4}\right]$	[9,11,30,32]
6	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{1}{4}\right]$	[10,12,29,31]
7	$\left[\frac{1}{2}, \frac{1}{2}-y, \frac{3}{4}\right]$	[13,15,26,28]
8	$\left[\frac{1}{2}, y+\frac{1}{2}, \frac{3}{4}\right]$	[14,16,25,27]

Table 11: Wyckoff site: 8k, site symmetry: m'm'2

No.	position	mapping
1	$[0, 0, z]$	[1,4,22,23]
2	$\left[0, 0, \frac{1}{2}-z\right]$	[2,3,21,24]
3	$[0, 0, -z]$	[5,8,18,19]
4	$\left[0, 0, z+\frac{1}{2}\right]$	[6,7,17,20]
5	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[9,12,30,31]
6	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[10,11,29,32]
7	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[13,16,26,27]
8	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[14,15,25,28]

Table 12: Wyckoff site: 8l, site symmetry: m'm'2

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,4,22,23]
2	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[2,3,21,24]
3	$\left[0, \frac{1}{2}, -z\right]$	[5,8,18,19]
4	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[6,7,17,20]
5	$\left[\frac{1}{2}, 0, z\right]$	[9,12,30,31]
6	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[10,11,29,32]
7	$\left[\frac{1}{2}, 0, -z\right]$	[13,16,26,27]
8	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[14,15,25,28]

Table 13: Wyckoff site: `16m`, site symmetry: `..2`

No.	position	mapping
1	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	`[1,12]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[2,11]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[3,10]`
4	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	`[4,9]`
5	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	`[5,16]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[6,15]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[7,14]`
8	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	`[8,13]`
9	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	`[17,28]`
10	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	`[18,27]`
11	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	`[19,26]`
12	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	`[20,25]`
13	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	`[21,32]`
14	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	`[22,31]`
15	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	`[23,30]`
16	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	`[24,29]`

Table 14: Wyckoff site: `16n`, site symmetry: `m'..`

No.	position	mapping
1	$[0,\ y,\ z]$	`[1,22]`
2	$\left[0,\ -y,\ \frac{1}{2}-z\right]$	`[2,21]`
3	$\left[0,\ y,\ \frac{1}{2}-z\right]$	`[3,24]`
4	$[0,\ -y,\ z]$	`[4,23]`
5	$[0,\ -y,\ -z]$	`[5,18]`
6	$\left[0,\ y,\ z+\frac{1}{2}\right]$	`[6,17]`
7	$\left[0,\ -y,\ z+\frac{1}{2}\right]$	`[7,20]`
8	$[0,\ y,\ -z]$	`[8,19]`
9	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z\right]$	`[9,30]`
10	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[10,29]`
11	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[11,32]`
12	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z\right]$	`[12,31]`
13	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ -z\right]$	`[13,26]`
14	$\left[\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[14,25]`
15	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	`[15,28]`
16	$\left[\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[16,27]`

Table 15: Wyckoff site: `16o`, site symmetry: `.m'.`

No.	position	mapping
1	$[x,\ 0,\ z]$	`[1,23]`
2	$\left[x,\ 0,\ \frac{1}{2}-z\right]$	`[2,24]`
3	$\left[-x,\ 0,\ \frac{1}{2}-z\right]$	`[3,21]`
4	$[-x,\ 0,\ z]$	`[4,22]`

continued ...

Table 15

No.	position	mapping
5	$[-x,\ 0,\ -z]$	[5,19]
6	$\left[-x,\ 0,\ z+\frac{1}{2}\right]$	[6,20]
7	$\left[x,\ 0,\ z+\frac{1}{2}\right]$	[7,17]
8	$[x,\ 0,\ -z]$	[8,18]
9	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z\right]$	[9,31]
10	$\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[10,32]
11	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[11,29]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z\right]$	[12,30]
13	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ -z\right]$	[13,27]
14	$\left[\frac{1}{2}-x,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[14,28]
15	$\left[x+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,25]
16	$\left[x+\frac{1}{2},\ \frac{1}{2},\ -z\right]$	[16,26]

Table 16: Wyckoff site: 16p, site symmetry: ..m

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,8]
2	$\left[x,\ -y,\ \frac{1}{2}\right]$	[2,7]
3	$\left[-x,\ y,\ \frac{1}{2}\right]$	[3,6]
4	$[-x,\ -y,\ 0]$	[4,5]
5	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[9,16]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[10,15]
7	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[11,14]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ 0\right]$	[12,13]
9	$\left[x,\ y,\ \frac{1}{2}\right]$	[17,24]
10	$[x,\ -y,\ 0]$	[18,23]
11	$[-x,\ y,\ 0]$	[19,22]
12	$\left[-x,\ -y,\ \frac{1}{2}\right]$	[20,21]
13	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[25,32]
14	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[26,31]
15	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[27,30]
16	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[28,29]

Table 17: Wyckoff site: 16q, site symmetry: ..m'

No.	position	mapping
1	$\left[x,\ y,\ \frac{1}{4}\right]$	[1,24]
2	$\left[x,\ -y,\ \frac{1}{4}\right]$	[2,23]
3	$\left[-x,\ y,\ \frac{1}{4}\right]$	[3,22]
4	$\left[-x,\ -y,\ \frac{1}{4}\right]$	[4,21]
5	$\left[-x,\ -y,\ \frac{3}{4}\right]$	[5,20]
6	$\left[-x,\ y,\ \frac{3}{4}\right]$	[6,19]
7	$\left[x,\ -y,\ \frac{3}{4}\right]$	[7,18]
8	$\left[x,\ y,\ \frac{3}{4}\right]$	[8,17]

continued ...

Table 17

No.	position	mapping
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[9,32]
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[10,31]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[11,30]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{4}\right]$	[12,29]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[13,28]
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[14,27]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{3}{4}\right]$	[15,26]
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[16,25]

Table 18: Wyckoff site: 32r, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[x,\, -y,\, \frac{1}{2}-z\right]$	[2]
3	$\left[-x,\, y,\, \frac{1}{2}-z\right]$	[3]
4	$[-x,\, -y,\, z]$	[4]
5	$[-x,\, -y,\, -z]$	[5]
6	$\left[-x,\, y,\, z+\frac{1}{2}\right]$	[6]
7	$\left[x,\, -y,\, z+\frac{1}{2}\right]$	[7]
8	$[x,\, y,\, -z]$	[8]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[9]
10	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[10]
11	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[12]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[13]
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14]
15	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[15]
16	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[16]
17	$\left[x,\, y,\, z+\frac{1}{2}\right]$	[17]
18	$[x,\, -y,\, -z]$	[18]
19	$[-x,\, y,\, -z]$	[19]
20	$\left[-x,\, -y,\, z+\frac{1}{2}\right]$	[20]
21	$\left[-x,\, -y,\, \frac{1}{2}-z\right]$	[21]
22	$[-x,\, y,\, z]$	[22]
23	$[x,\, -y,\, z]$	[23]
24	$\left[x,\, y,\, \frac{1}{2}-z\right]$	[24]
25	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[25]
26	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[27]
28	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[28]
29	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[29]
30	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[30]
31	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[31]
32	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, \frac{1}{2}-z\right]$	[32]